ASSIGNMENT 3 / EXPERIMENT SUMMARY

Forecasting tomorrow's temperature

I compared ten forecasting approaches on the Jena weather dataset.

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VALIDATION-SELECTED WINNER

GRU 32

Lowest validation error among all ten approaches

2.296°C

Held-out test MAE

BASELINE (VALIDATION)

2.586°C

Same temperature 24h later

BEST (VALIDATION)

2.380°C

GRU with 32 units

ERROR REDUCTION

8.0%

Compared with baseline

THE MAIN FINDING

I selected the 32-unit GRU because it had the lowest validation MAE. On the held-out test, it reached 2.296°C, compared with 2.500°C for the baseline.

The stacked GRU 64-32 had almost the same test result. A gap of just 0.000083°C in one run is too small to establish that either model is generally better.

SOURCE: EXECUTED KAGGLE NOTEBOOK + VERIFIED PREDICTIONS

The question and the data

I tested whether five days of weather observations could predict air temperature 24 hours ahead.

RAW READINGS	HOURLY READINGS	WEATHER FEATURES
420,551	70,129	14

PREPARATION

01 Original station data

The dataset records temperature and 13 other weather measurements every ten minutes.

02 Hourly view

I averaged the observations by hour, replaced 38 invalid markers, and filled 88 missing temperature hours using prior values.

03 Time-based holdouts

I used older observations for training, later observations for validation, and the newest data for testing.

04 Training-only normalization

I fitted input scaling on the training period to avoid using information from the future.

CHRONOLOGICAL SPLIT

SET	HOURLY ROWS	PERIOD
Training	49,090	Jan 2009 - Aug 2014
Validation	10,519	Aug 2014 - Oct 2015
Test	10,520	Oct 2015 - Jan 2017

ONE FORECAST WINDOW

Each window uses 120 hourly readings (5 days) to predict temperature 24 hours ahead. I started a new window every 6 hours, yielding 1,730 final test examples.

All ten approaches, one evaluation rule

I ranked the models by validation MAE. I report their test results for completeness but did not use them to change the winner.

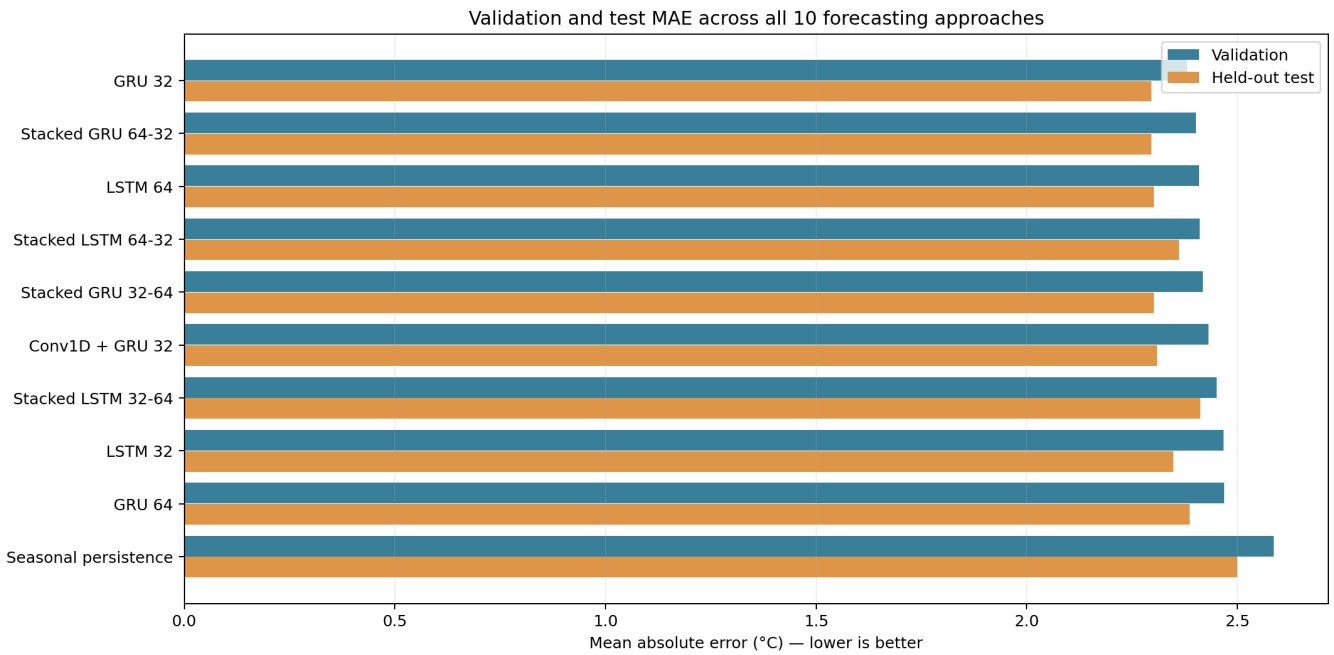


TABLE 1 / MODEL COMPARISON

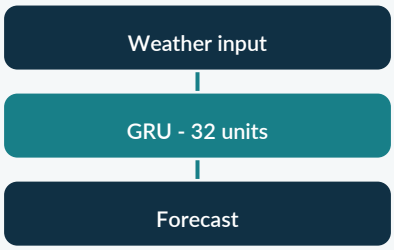
MODEL (VALIDATION RANK)	VAL MAE	TEST MAE
01 GRU 32	2.380	2.296
02 Stacked GRU 64-32	2.401	2.296
03 LSTM 64	2.409	2.302
04 Stacked LSTM 64-32	2.411	2.361
05 Stacked GRU 32-64	2.417	2.302
06 Conv1D + GRU 32	2.431	2.309
07 Stacked LSTM 32-64	2.450	2.413
08 LSTM 32	2.467	2.347
09 GRU 64	2.469	2.387
10 Seasonal persistence	2.586	2.500

All nine neural models had lower MAE than the persistence baseline on validation and held-out test data. Lower MAE is better; values are in °C.

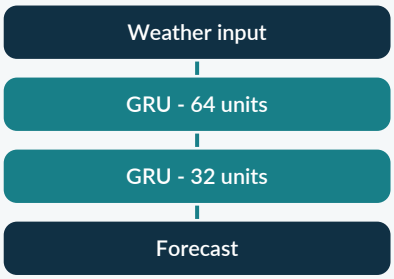
Did adding recurrent layers help?

I tested model depth and unit order alongside the single-layer GRU and LSTM models.

SINGLE-LAYER GRU



STACKED GRU



STACKED MODEL RESULTS

CONFIGURATION	VALIDATION	TEST
Stacked GRU 64 → 32	2.401°C	2.296°C
Stacked LSTM 64 → 32	2.411°C	2.361°C
Stacked GRU 32 → 64	2.417°C	2.302°C
Stacked LSTM 32 → 64	2.450°C	2.413°C

What was learned

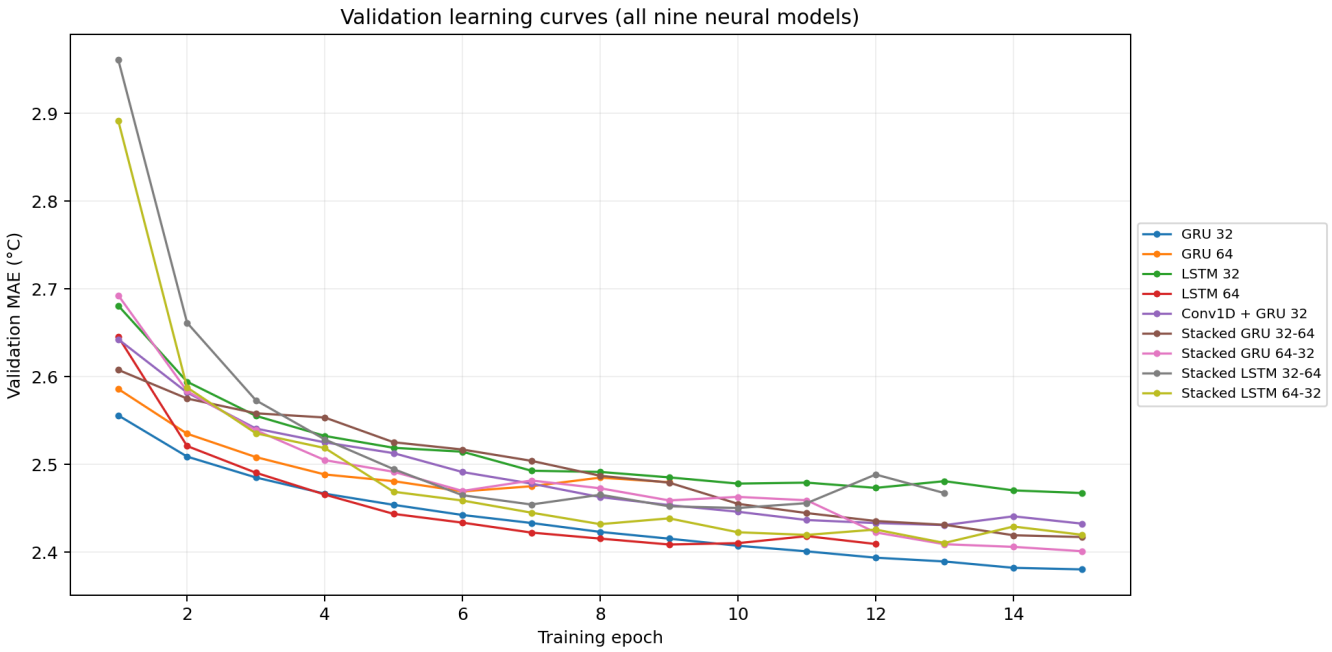
I tried both unit orders (32 to 64 and 64 to 32) for GRU and LSTM. GRU 64 to 32 was the best stacked model on validation, but the single-layer GRU 32 still did better.

A difference too small to overinterpret

The stacked GRU 64 to 32 had a test MAE of 2.295876°C; GRU 32 reached 2.295959°C. That 0.000083°C gap does not establish an advantage for stacking.

How the networks learned

I used the same training budget and early-stopping rule for every neural architecture.



TRAINING SETTINGS

Training budget	Up to 15 epochs
Batch size	256 forecast windows
Optimization	Adam; learning rate 0.001; MAE loss
Early stopping	Patience 3, best weights restored
Selection rule	Lowest validation MAE, not lowest test MAE

Interpretation

Some curves were still improving near epoch 15, while early stopping ended other runs sooner. This is one fixed-seed run, not a full hyperparameter search.

What happened on unseen weather?

I selected the winner on validation data before testing every approach on the same later weather period.

WINNER TEST MAE

2.296°C

GRU 32; 1,730 examples

BASELINE TEST MAE

2.500°C

24-hour persistence

EXAMPLE HELD-OUT FORECAST

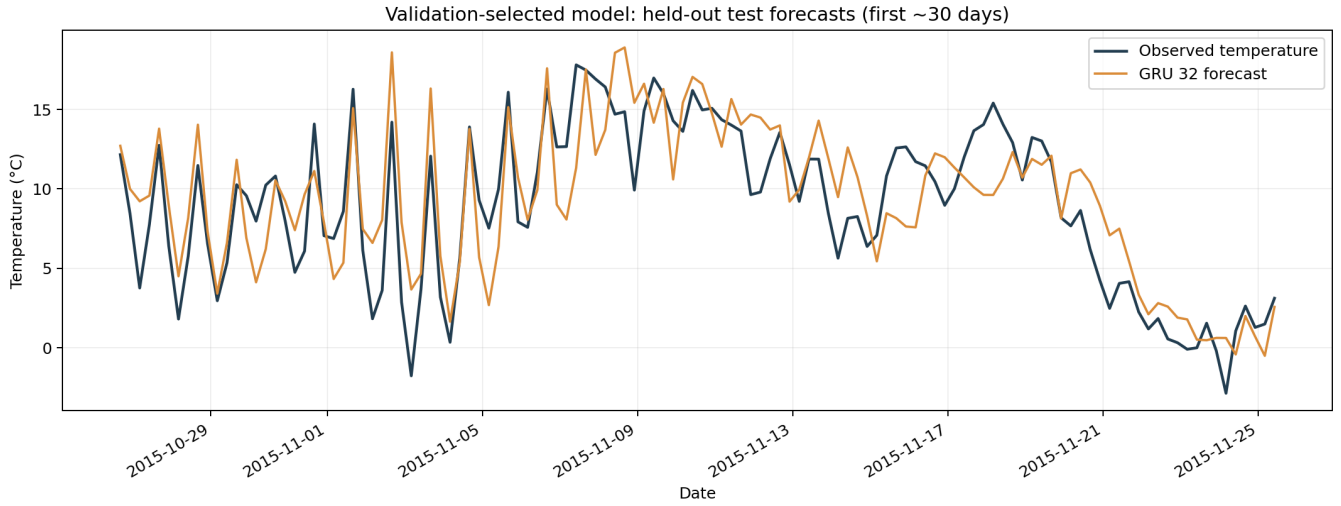


Figure 3. The first 120 held-out forecasts span about 30 days, with a new prediction every six hours. The chart compares observed temperatures with GRU 32 predictions.

How to interpret MAE

A test MAE of 2.296°C means the selected GRU missed the observed temperature by about 2.30°C on average. Individual errors vary. The average test error was 8.2% lower than the baseline.

I report test MAE for all ten approaches, but kept GRU 32 as the winner because I did not use test results to revise the validation-based selection.

What the experiments tell us

The recurrent models improved forecasts on this dataset, though the results come from a limited experiment.

01

Recurrent forecasting improved the baseline

All nine neural models beat the persistence baseline on validation and test data. GRU 32 reduced validation error by 8.0% and test error by 8.2%.

02

GRU 32 was the validation winner

GRU 32 reached 2.380°C validation MAE and 2.296°C test MAE across 1,730 forecasts.

03

Stacking was explored but not decisive

GRU 64 to 32 was the best stacked model on validation (2.401°C). It nearly tied GRU 32 on the test, but did not beat it on validation.

04

More units or complexity did not guarantee gains

Increasing GRU width from 32 to 64 raised validation MAE. Neither Conv1D + GRU nor the stacked LSTM models beat the best single-layer GRU.

05

Model ranking is specific to this run

This ranking comes from one fixed seed, one dataset, one chronological split and limited tuning. Small differences could change in another run.

ASSIGNMENT COVERAGE

I tested recurrent forecasts, different unit counts, stacked GRU and LSTM layers, LSTM in place of GRU, and Conv1D with a recurrent layer. I compared validation results and reported test MAE for every configuration.

GRU 32 was the strongest model selected on validation in this experiment. Greater depth or width did not consistently improve accuracy.